



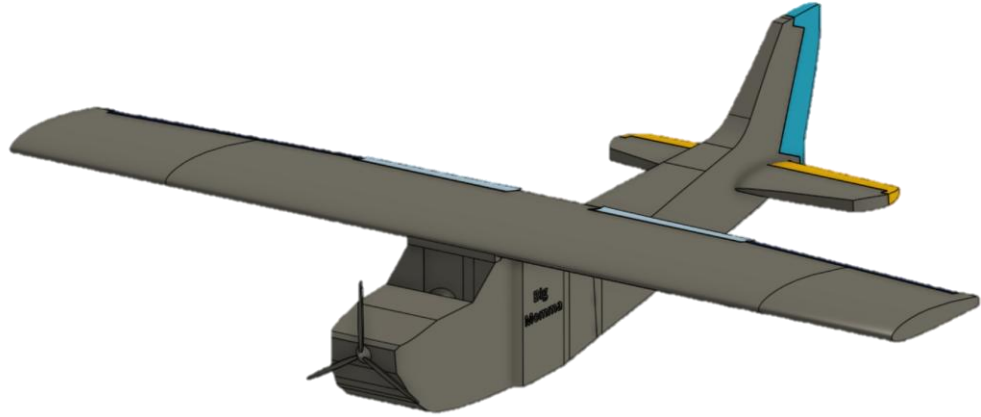
Aircraft Design and Flight Dynamics Project

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Description of Design Process

- Given project manual constraints (4 passengers and small fixed-wing aircraft) and some initial research, we chose to model our aircraft similarly to the Cessna 172 Skyhawk
- In order to create the model, top down, isometric, and side view drawings of the aircraft were obtained and uploaded into OnShape to be able to accurately design a similar aircraft.
- After making an initial model, without control surfaces, for the first report, we decided to hollow out the wings and fuselage of the aircraft so that fuel, engine, and passenger weight could be accurately represented
- Material selection and weight distribution was calculated using OnShape's material library for the skin (Al6061) as well as creating custom materials to match typical fuel weight and required engine/passenger weights.
- For the second report, we used inspiration for the Cessna 172 Skyhawk as well as other Cessna aircrafts to design the control surfaces such that the dynamic stability of our model aligned with industry standards

3D Model

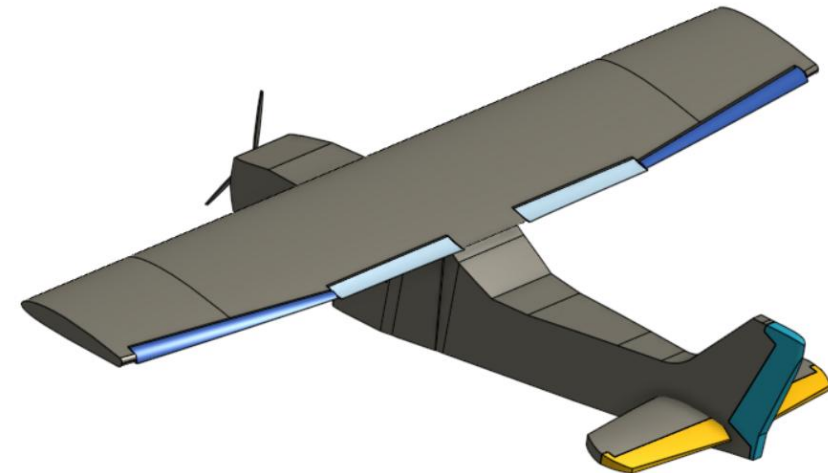
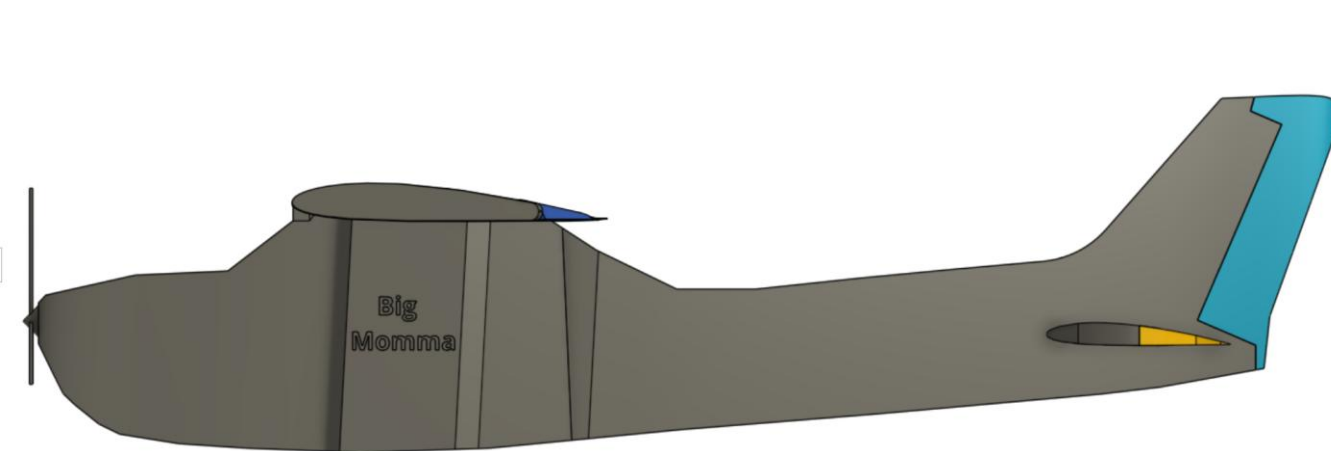


Modeled using OnShape

The four passengers are in the body of the plane.

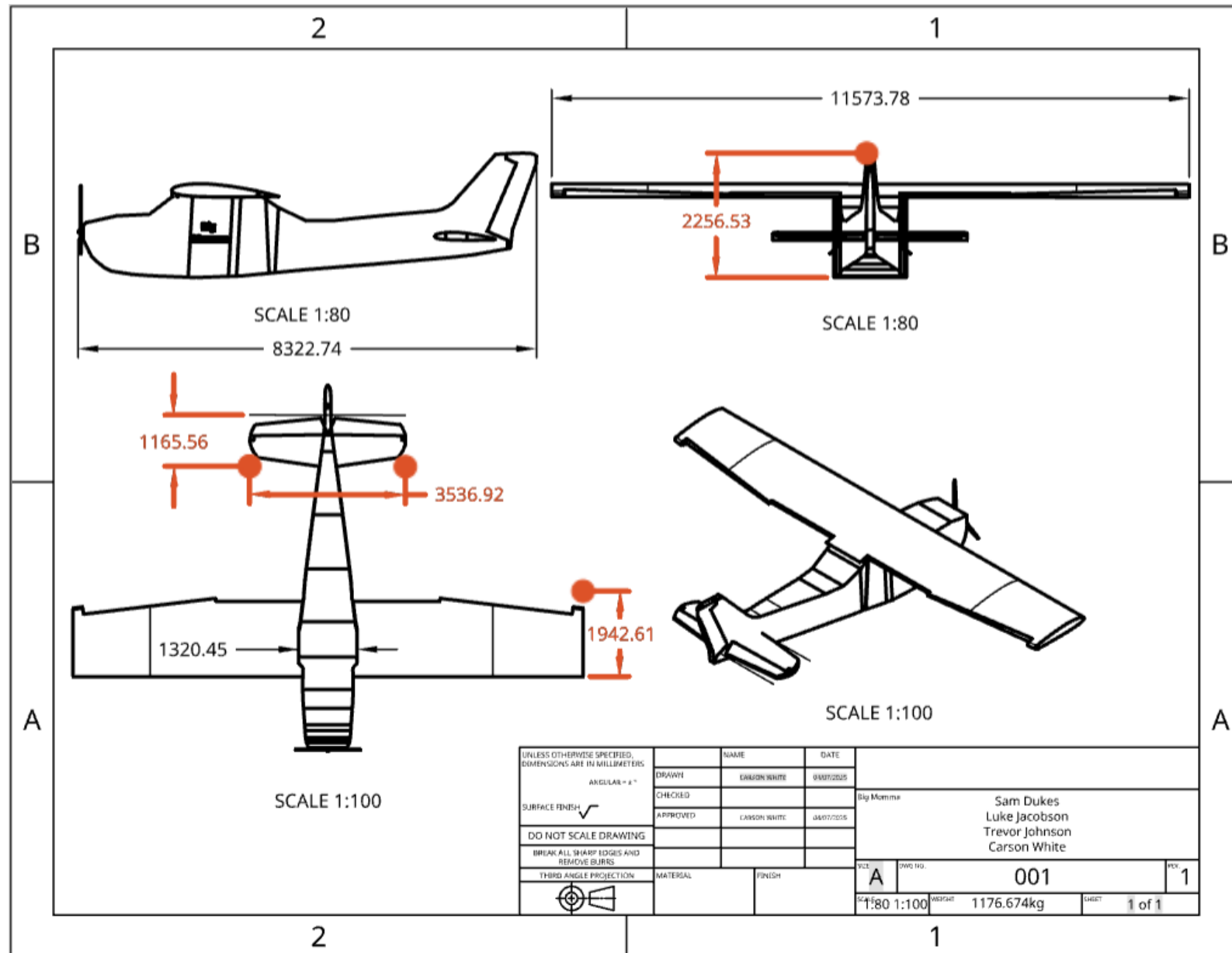
The engine is positioned in front of the cockpit.

The fuel is distributed throughout each side of the wing for stability. Ailerons are Kentucky blue, flaps are baby blue, elevators are mustard yellow, and rudder is turquoise.



Key Dimensions

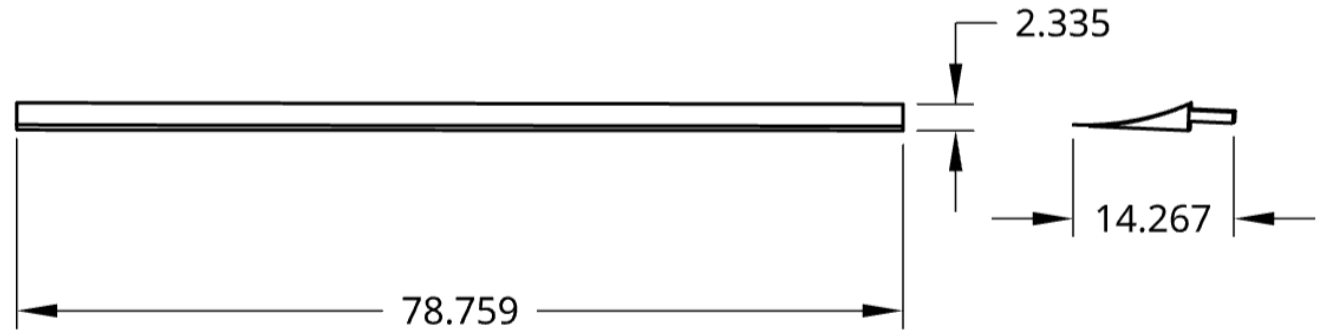
*Units in millimeters



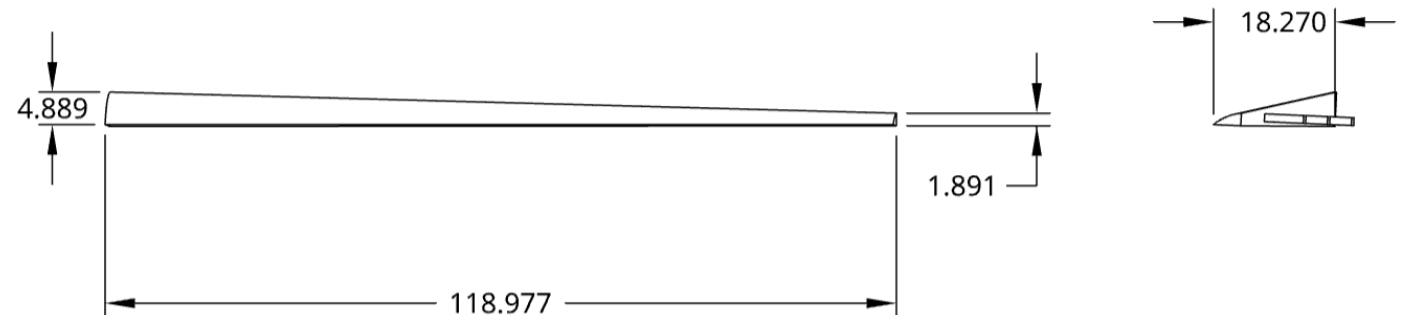
Key Dimensions: Control Surfaces of wing

*Units in millimeters

Flaps:



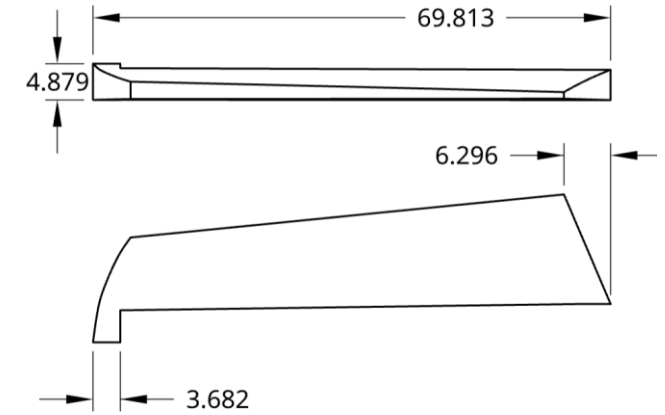
Ailerons:



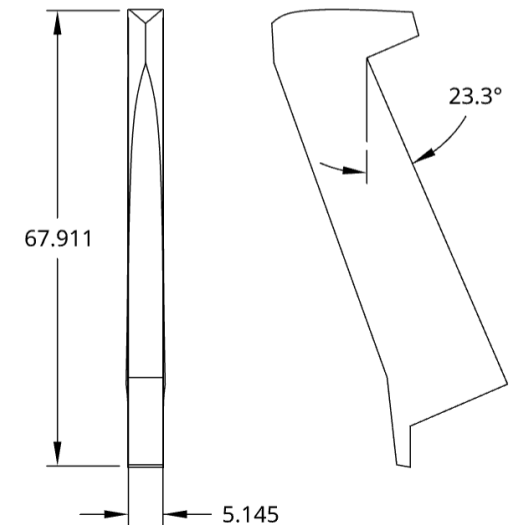
Key Dimensions: Control Surfaces of stabilizers

*Units in millimeters

Elevators:

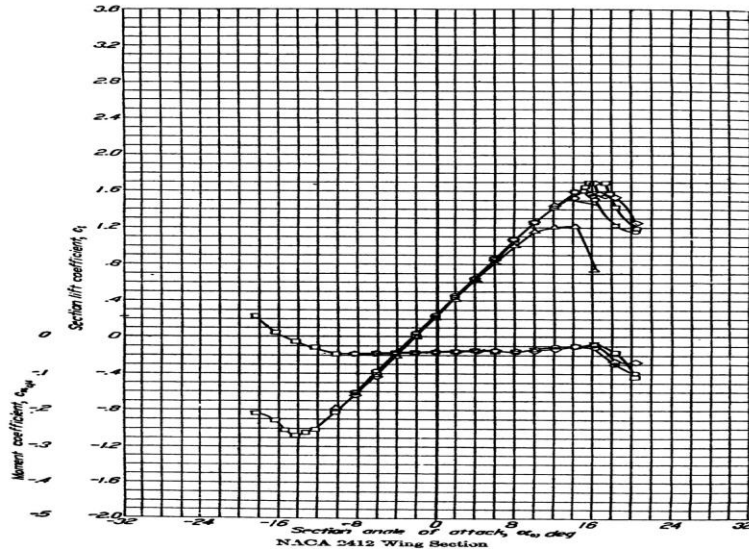


Rudder:



NACA Airfoil

The wing was designed using the **NACA 2412** and the horizontal stabilizer used the **NACA 0012**



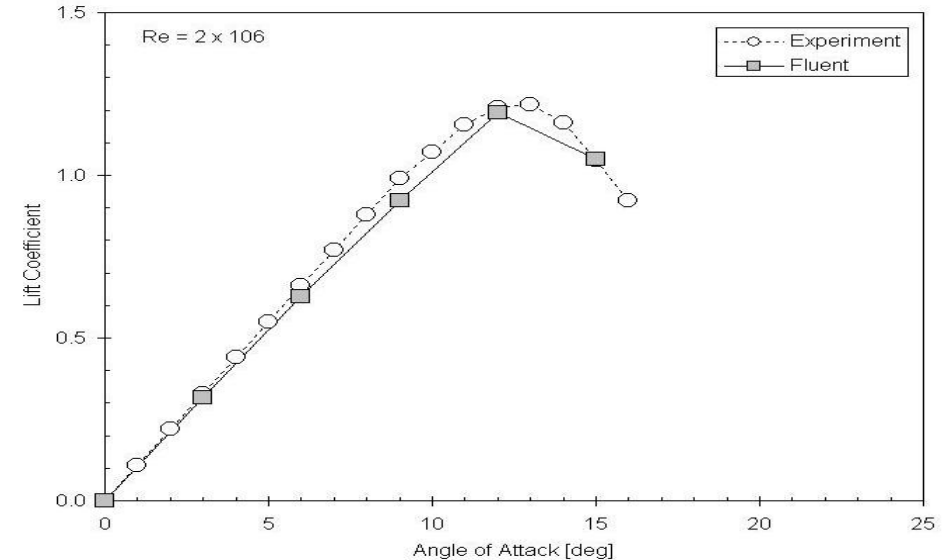
Wing Reference Dimensions:

$$C = 1.945 \text{ m}$$

$$b = 11.574 \text{ m}$$

$$S = 22.511 \text{ m}^2$$

$$X_{AC_{wing}} = 6.134 \text{ m}$$



Horizontal Stabilizer Reference Dimensions:

$$C = 1.167 \text{ m}$$

$$b = 3.537 \text{ m}$$

$$S = 4.128 \text{ m}^2$$

$$X_{AC_{tail}} = 1.553 \text{ m}$$

Key Design Points

Values obtained using Onshape's "Mass and Section Properties" tool:

Mass	☒ Override				1161.371 kg
Volume					11.47 m ³
Surface area					84.852 m ²
Center of mass	☒ Override				
X 					0.001 m
Y 					5.444 m
Z 					-1.281 m
Mass moments of inertia (kg m ²)	☒ Override				
Lxx	2587.461	Lxy	-0.767	Lxz	1.454
Lyx	-0.767	Lyy	2953.51	Lyz	-16.851
Lzx	1.454	Lzy	-16.851	Lzz	5123.23

*The origin is located at the top end of the vertical stabilizer

Mass Matrix AVL Input:

mass	x	y	z	lxx	lyy	lzz	lxy	lxz	lyz
1161.371	-5.444	0.001	1.281	2953.51	2587.461	5123.23	0.767	16.851	1.454

Static Longitudinal Stability: Methodology

To determine if our model has static longitudinal stability, we analyzed the pitching moment due to a perturbation in $\delta\alpha$, ensuring that $M_{A\alpha} < 0$, which is the condition required for static stability. This relation,

$$\frac{\partial L}{\partial \alpha} = \frac{1}{2} \rho V^2 S C_{L\alpha}$$

where ρ , V , and $C_{L\alpha}$ are equivalent for the tail and wing, was used to simplify our analysis. This led us to find that

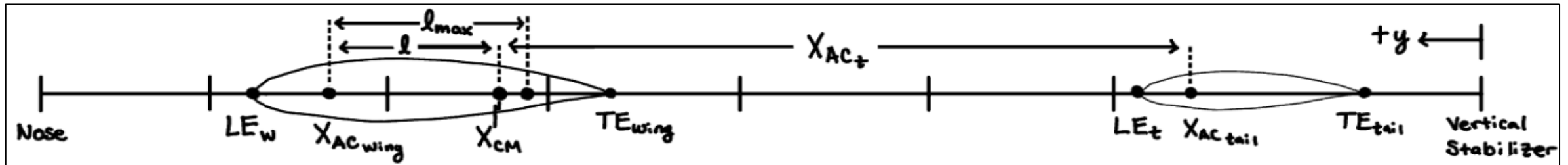
$$l_{max} = X_{AC_t} \frac{\left. \frac{\partial L_t}{\partial \delta\alpha} \right|_0}{\left. \frac{\partial L_w}{\partial \delta\alpha} \right|_0} = X_{AC_t} \frac{S_t}{S_w}$$

where l_{max} is the maximum distance the aerodynamic center of the wing ($X_{AC_{wing}}$) can be in front of the center of mass of the plane (X_{CM}). X_{AC_t} is the distance from X_{CM} to $X_{AC_{tail}}$. S_t and S_w are the planform areas of the tail and wing respectively. The aerodynamic centers were assumed to be $\approx 25\%$ of the chord length.

Static Longitudinal Stability: Results

We found l_{max} to be $0.71m$. If the distance l (X_{CM} to $X_{AC,wing}$) is greater than l_{max} , the model will be unstable.

Since $l = 0.708m$, our model has static longitudinal stability.

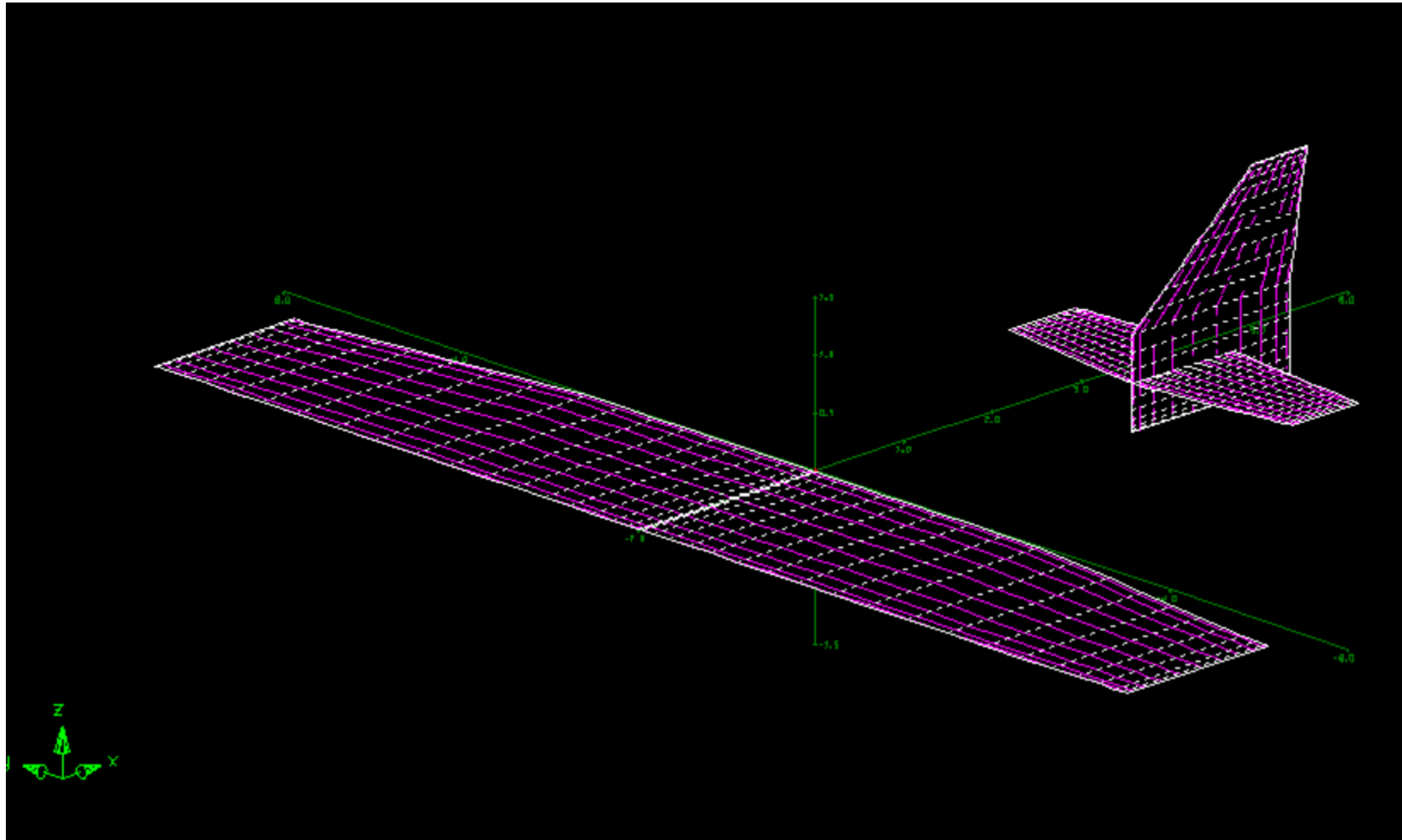


Critical Distances:

	Nose	LE_w	$X_{AC,wing}$	X_{CM}	TE_w	LE_t	$X_{AC,tail}$	TE_t	V.S.	l	l_{max}	$X_{AC,t}$
y [m]	8.323	6.62	6.134	5.426	4.681	1.845	1.553	0.68	0	0.708	0.71	3.873

*Variable definitions: W (wing), t (tail), LE (leading edge of airfoil), TE (trailing edge of airfoil), CM (center of mass), $V.S.$ (vertical stabilizer)

AVL Model



Steady Conditions

Standard axis orientation, X fwd, Z down

Run case: stability derivatives

Alpha =	0.86136	pb/2V =	-0.00000	p'b/2V =	-0.00000
Beta =	0.00000	qc/2V =	-0.00000		
Mach =	0.231	rb/2V =	-0.00000	r'b/2V =	-0.00000
CXtot =	-0.01493	Cltot =	-0.00000	Cl'tot =	-0.00000
CYtot =	0.00000	Cmtot =	-0.00000		
CZtot =	-0.16124	Cntot =	-0.00000	Cn'tot =	-0.00000
CLtot =	0.16100				
CDtot =	0.01736				
CDvis =	0.00000	CDind =	0.0173555		
CLff =	0.16126	CDff =	0.0175574	Trefftz	
CYff =	0.00000	e =	0.0792	Plane	
Rudder	=	0.00000			
flap	=	0.00000			
aileron	=	0.00000			
elevator	=	-26.57529			

$$\alpha_0 = 0.86136^\circ$$

$$\delta e_0 = -26.57^\circ$$

$$CL_0 = 0.161$$

Stability Derivatives

Geometry-axis derivatives...

	axial	vel. u	sideslip	vel. v	normal	vel. w		
x force CX	CXu	= -0.032543	CXv	= -0.000000	CXw	= 0.177774		
y force CY	CYu	= 0.000000	CYv	= -0.235550	CYw	= -0.000000		
z force CZ	CZu	= -0.259535	CZv	= 0.000000	CZw	= -4.189470		
x mom. Cl	Clu	= -0.000000	Clv	= 0.009897	Clw	= 0.000000		
y mom. Cm	Cmu	= 0.136277	Cmv	= 0.000000	Cmw	= -9.064187		
z mom. Cn	Cnu	= -0.000000	Cnv	= 0.196184	Cnw	= -0.000000		
	roll rate p		pitch rate q		yaw rate r			
x force CX	CXp	= 0.000000	CXq	= 0.465847	CXr	= -0.000000		
y force CY	CYp	= 0.099590	CYq	= -0.000000	CYr	= 0.428912		
z force CZ	CZp	= 0.000000	CZq	= -23.827361	CZr	= 0.000000		
x mom. Cl	Clp	= -0.362487	Clq	= 0.000000	Clr	= 0.041644		
y mom. Cm	Cmp	= 0.000000	Cmq	= -57.600556	Cmr	= 0.000000		
z mom. Cn	Cnp	= -0.080175	Cnq	= 0.000000	Cnr	= -0.362519		
	Rudder	d01	flap	d02	aileron	d03	elevator	d04
x force CX	CXd01	= 0.000000	CXd02	= 0.000066	CXd03	= -0.000000	CXd04	= 0.000569
y force CY	CYd01	= -0.003243	CYd02	= 0.000000	CYd03	= 0.000293	CYd04	= -0.000000
z force CZ	CZd01	= -0.000000	CZd02	= -0.016473	CZd03	= 0.000000	CZd04	= -0.005911
x mom. Cl	Cl d01	= 0.000317	Cl d02	= 0.000000	Cl d03	= -0.005822	Cl d04	= 0.000000
y mom. Cm	Cm d01	= -0.000000	Cm d02	= -0.033126	Cm d03	= -0.000000	Cm d04	= -0.027591
z mom. Cn	Cn d01	= 0.002696	Cn d02	= -0.000000	Cn d03	= 0.000054	Cn d04	= 0.000000

$C_{M_w} \cong C_{M_\alpha} < 0 = -9.06$ Static Longitudinal Stability
 $C_{l_v} \cong C_{l_\beta} < 0 = 0.19$ Not Static Lateral Stability
 $C_{n_v} \cong C_{n_\beta} > 0 = 0.009897$ Static Directional Stability

Equations of Motion

From AVL's mode function:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \\ \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} -0.0257 & 0.14 & -0.8253 & -9.81 & 0 & 0 \\ -0.1996 & -3.2224 & 57.182 & 0 & 0 & 0 \\ 0.0905 & -6.0845 & -37.6029 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1.148 & 0 & 0 \\ 0 & 1 & 0 & -77.1615 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \\ x \\ z \end{bmatrix} + \begin{bmatrix} 4.10\text{E-}17 & 0.003962 & -3.56\text{E-}17 & 0.03464 \\ -9.33\text{E-}17 & -0.9778 & -6.52\text{E-}17 & -0.3508 \\ -0.00026 & -1.716 & 0.000423 & -1.429 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ \delta e \end{bmatrix}^T$$

From calculation using stability derivatives and mass matrix:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -0.0257 & 0.140387 & 1.391682 & -9.81 \\ -0.20495 & -3.30839 & 40.56235 & 0 \\ 0.09395 & -6.24887 & -77.2358 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} 0.004021631 \\ -1.00376259 \\ -1.76214634 \\ 0 \end{bmatrix} \begin{bmatrix} \delta e \end{bmatrix}^T$$

Natural Frequency & Dampening Ratios

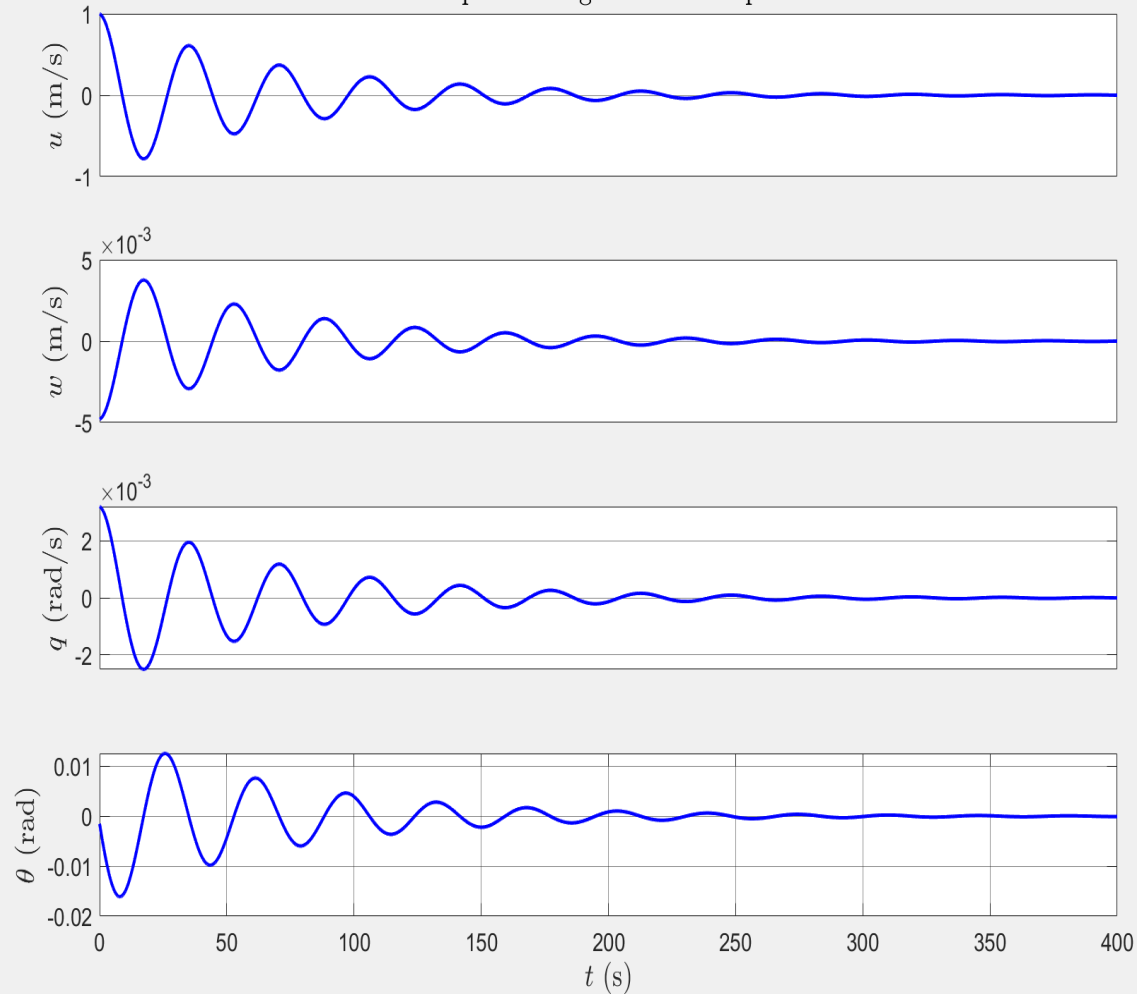
	Run case	Eigenvalue	
Short Period	1	-20.372706	7.2892280
	1	-20.372706	-7.2892280
Dutch Roll	1	-2.3269832	5.0869560
	1	-2.3269832	-5.0869560
Roll Subsidence	1	-6.7363911	0.0000000
Spiral	1	0.21267183E-01	0.0000000
Phugoid	1	-0.14110268E-01	0.17692460
	1	-0.14110268E-01	-0.17692460

Mode	Natural Frequency (ω_n)	Damping Ratio (ζ)
Short-Period	21.6375 [rad/s]	0.9415
Phugoid	0.1775 [rad/s]	0.0800

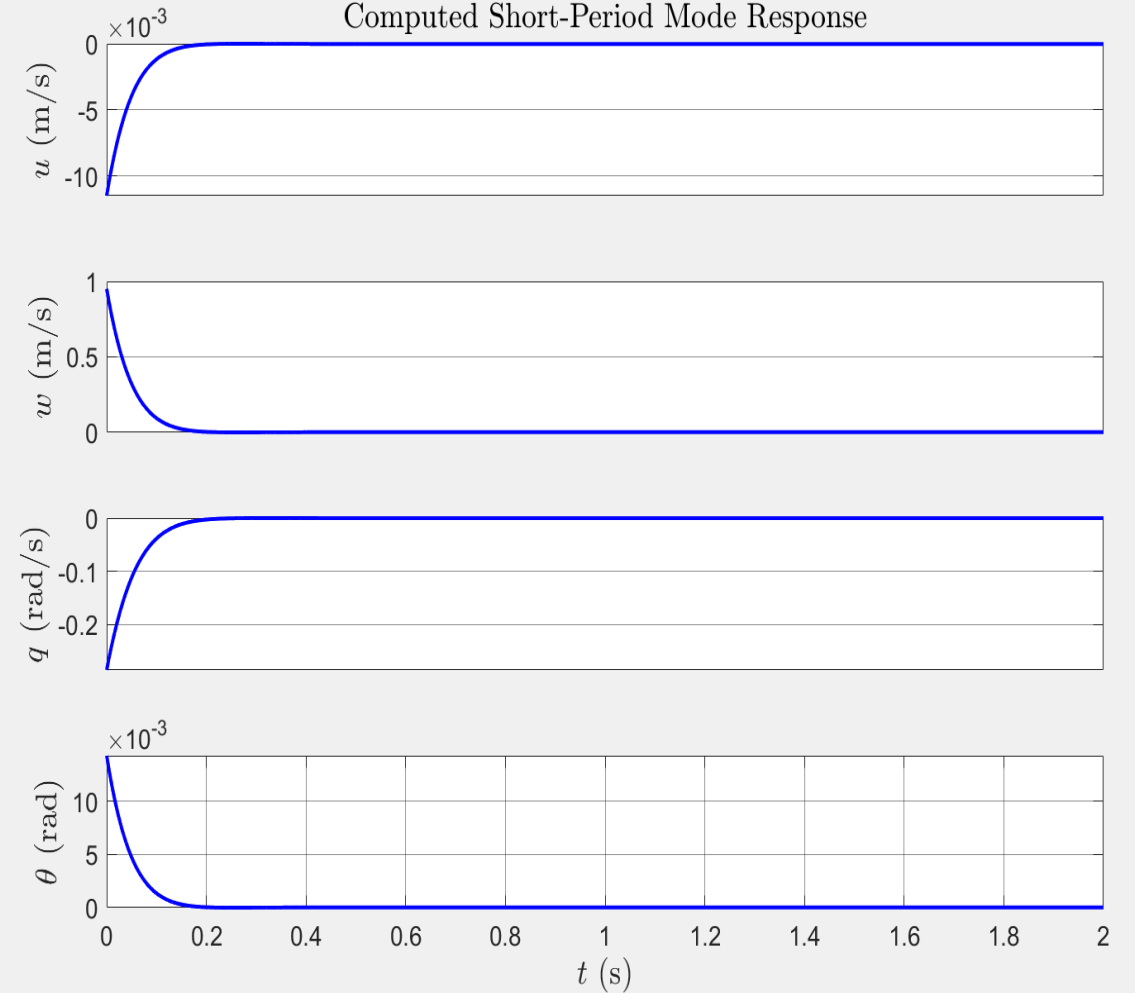
For a Category A, B, and C flight phase, our calculated damping ratios fall within the Level 1 flying qualities region. This indicates that the aircraft requires minimal pilot input to maintain steady flight, as it naturally returns to equilibrium with little effort.

Numerical Simulation of Flight Modes

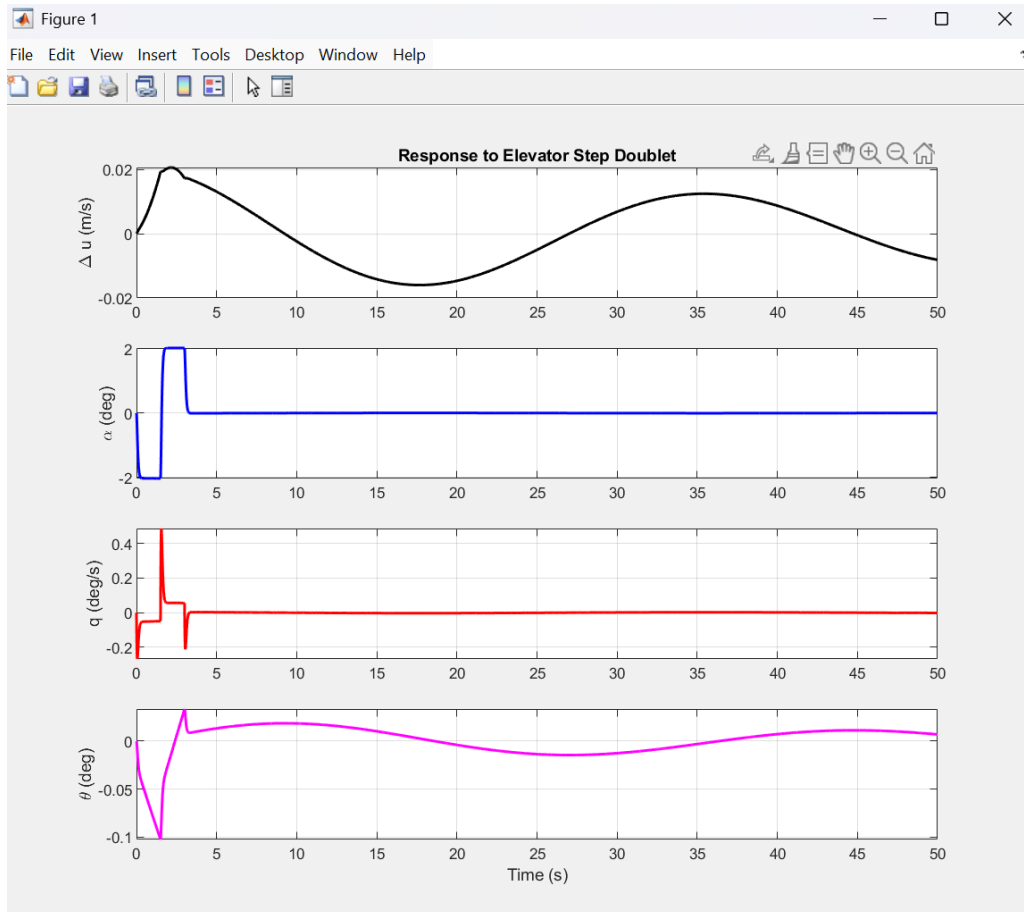
Computed Phugoid Mode Response



Computed Short-Period Mode Response



Numerical Simulation of Elevator Step-Doublet Response



Parameter: 10-degree deflection at 1.5 seconds each then returns to 0 degrees of deflection

Future Considerations

- Incorporate a wing dihedral to gain lateral static stability
- Further analyze the discrepancies between AVL and the calculated values of the state space system
- Decrease the steady state deflection angle by increasing the span of the horizontal stabilizer (changing this variable in AVL did decrease the deflection angle to a normal value)

Appendix: Detailed CAD Model

Ailerons/Flaps:



Chrome PDF
Document

Aircraft without control surfaces:



Chrome PDF
Document

Vertical/Horizontal Stabilizers:



Chrome PDF
Document



Compressed
(zipped) Folder

Appendix: AVL Files

Input Files:



Geometry_File.avl



Mass_Files.mass



Runcase_File.txt

Output Files:



AVL_State-Space_Matrix.txt



Stability_Derivatives_and_Steady_Conditions.txt



AVL_Eigenvalues_Output.txt

Appendix: Matlab Code

- For Doublet Response:



- For Phugoid Solution:



- For Short-Period Solution:

